Why distributional pressure buys nuisance

A linear theory of JEPA regularisers and its test

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Abstract A companion study of regularised joint-embedding predictive architectures (JEPAs) on 3DShapes left three findings unexplained: nuisance leakage rose with the regularisation weight λ for every regulariser, noise on the predictor input raised leakage, and a non-Gaussian prior matched to the content law did not improve identifiability. We derive a small theory in a linear model with k content and m nuisance dimensions and a d-dimensional embedding, and test it on the same data and code. With a covariance penalty we prove that nuisance enters only above a threshold $\lambda^* = c_p\,\alpha(2-\alpha)/2$ for a learned linear predictor (α is the nuisance resampling rate), not $\lambda^* \propto \alpha$ as first sketched. A linear probe's leakage is a step at λ^*, and directions that neither content nor nuisance can fill stay collapsed. For any regulariser that depends only on the embedding covariance, the threshold is $c_p\,\alpha(2-\alpha)/t^*_{\mathrm{reg}}$, with a per-regulariser constant that we compute numerically. We registered 18 predictions before training any network and ran 1075 runs on 5 held-out seeds. The theory predicts the order of the thresholds (SIGReg < KSD < sliced KSD < MMD) and that they are not proportional to α (ratios 0.50 at α = 0.25, against 0.25). With the convolutional encoder the measured thresholds come at 0.29 times the predicted values (always early; with a linear pixel encoder, late). The theory fails at $d = k$, where leakage still appears, as it does when the same training loop runs the linear model. Linear leakage is also a transient: by 8000 steps it falls in all four tested cells, to about 0 in two, where the network fills the spare directions with nonlinear content features, a route our corollary shows to be free for the covariance penalty. At the higher λ levels nuisance stays decodable by kNN. Compared along their whole λ paths instead of at one tuned λ, the regularisers rank differently from the companion study's verdict. At the uniform law and study 1′s λ, a matched prior raises a permutation-level metric (MCC) while orthogonal-fit R^2 is unchanged, which the theory allows, but the gain is not confined to where the theory placed it; at the Laplace law the matched prior lowers both.

1 Introduction

JEPAs predict one view's embedding from another's and need a second ingredient to avoid a constant embedding [1]. One family regularises the batch of embeddings towards a target law: VICReg constrains variances and covariances [2], SIGReg tests one-dimensional projections against a standard Gaussian [3], and kernel discrepancies (MMD, kernel Stein discrepancy (KSD), sliced KSD) compare the batch with $\mathcal{N}(0, I)$ [4], [5], [6]. A companion study [7] compared these regularisers on 3DShapes [8], with the generative factors split into content (object hue, scale, shape; a $k = 6$ dimensional code) and nuisance (floor hue, wall hue, orientation; a 5-dimensional code) that is resampled between views with probability α. Leakage was the held-out R^2 of a linear probe for nuisance. Five of its six hypotheses were refuted, and three findings were left unexplained:

1. **Leakage follows λ, not the target law.** For every regulariser, VICReg included, leakage rose with the weight λ, often abruptly (KSD at $d = 12$: 0.002 at λ = 0.2, 0.382 at λ = 1, 0.533 at λ = 5). Which arm "leaked least" depended on the tuning rule.
2. **Noise on the predictor input raised leakage** (KSD, σ = 1: +0.646 [0.596, 0.687]), while narrowing d lowered it.
3. **A matched non-Gaussian prior did not improve identifiability**, measured by R^2 after an orthogonal fit.

This paper explains these findings with a linear theory and tests it. **Contributions.** (i) In a linear content/nuisance model we prove that a covariance penalty admits nuisance only above a threshold $\lambda^* = c_p\kappa/2$, where κ is the invariance cost per unit of nuisance variance: $\kappa = 2\alpha$ for an identity predictor and $\kappa = \alpha(2-\alpha)$ for a learned linear predictor (Theorem 1). We also show that linear-probe leakage is a step at λ^*, that directions nobody can fill stay collapsed, and that the threshold needs a linear encoder (Corollary 1). (ii) For any regulariser that depends on the embedding only through its covariance, the threshold is $c_p\kappa/t^*_{\mathrm{reg}}$ (Theorem 2). We compute t^*_{reg} numerically for SIGReg, KSD, sliced KSD and MMD and find that it is set by the regulariser's slope at a collapsed direction, not by its curvature at I. (iii) A heuristic analysis of predictor-input noise (Heuristic 2), and a proved invariance result on what a Gaussian-target arm can identify (Proposition 3), with the matched-prior case taken from Comon's ICA theorem [9]. (iv) 18 predictions registered before any network was trained, tested in 1075 runs on 5 held-out seeds, with arms compared along their whole λ path. Each prediction's outcome is reported, including the failures (Table 2).

2 Model and theory

Model. Content $s \in \mathbb{R}^k$ and nuisance $n \in \mathbb{R}^m$ are independent, zero mean, with identity covariance. A pair of views is $x = (s, n)$, $x' = (s, n')$, where each nuisance coordinate is resampled independently with probability $\alpha \in (0, 1]$, so $\mathrm{Cov}(n, n') = (1-\alpha)I$. The encoder is linear, $z = As + Bn \in \mathbb{R}^d$. Write $P = AA^\top$, $Q = BB^\top$, $\Sigma = P + Q$ for the embedding covariance, and $r = \min(m, (d-k)_+)$. The objective is $J = c_p L_{\mathrm{pred}} + \lambda R(\Sigma)$, with $L_{\mathrm{pred}} = \mathbb{E}\|z - z'\|^2$ (identity predictor) or $\min_W \mathbb{E}\|Wz - z'\|^2$ (learned linear predictor, trained jointly without stop-gradient, as in the companion study's regularised arms). In the code the prediction loss is a mean over the d coordinates, so $c_p = 1/d$; all λ values below are in code units. Study 1's sampler has this structure with $k = 6$ and $m = 5$ (it resamples whole factors, which keeps the $(1-\alpha)$ structure for each factor's features), so $m < d - k$ at $d = 12$ and $d = 24$.

Status labels. A *theorem*, *lemma*, *proposition* or *corollary* has a complete proof in Appendix A that was also confirmed numerically (gradient descent on the linear model from several starts, compared with the closed form; scripts `theory/check_t*.py`, tolerance 0.002 on population quantities). *Numerical* means that an exact reduction is proved and the final constant is computed. *Heuristic* means that it is not proved.

Lemma 1 (invariance cost). (i) $\mathbb{E}\|z - z'\|^2 = 2\alpha \operatorname{tr} Q$. (ii) $\min_W \mathbb{E}\|Wz - z'\|^2 = 2\alpha \operatorname{tr} Q - \alpha^2 \operatorname{tr}(Q\Sigma^+Q) \geq \alpha(2-\alpha)\operatorname{tr} Q$, with equality iff content and nuisance occupy linearly independent ranges ($P\Sigma^+Q = 0$).

Theorem 1 (covariance penalty; corrected and proved). Let $R = \|\Sigma - I\|_F^2$, κ as in Lemma 1, $\lambda^* = c_p\kappa/2$ and $v^* = (1 - \lambda^*/\lambda)_+$. (a) If $d \leq k$, every global minimiser has $B = 0$ and $AA^\top = I_d$. (b) If $d > k$, every global minimiser has $AA^\top = UU^\top$ for a k-dimensional orthonormal U. If $\lambda > \lambda^*$, then $BB^\top = v^*VV^\top$ with V an r-dimensional orthonormal set orthogonal to U; if $\lambda \leq \lambda^*$, then $B = 0$. In both cases the remaining $d - k - r$ directions have zero variance.

So $\lambda^* = c_p\alpha$ for the identity predictor, as first sketched, but $\lambda^* = c_p\alpha(2-\alpha)/2$ for the learned linear predictor, which is *not* proportional to α. When $m < d - k$, the $d - k - m$ directions that neither content nor nuisance can fill stay collapsed at every λ.

Corollary 1. (a) The population linear-probe R^2 of n from z is r/m for $\lambda > \lambda^*$ and 0 otherwise: it is scale-invariant, so it *steps* at λ^*. The nuisance share of embedding variance, $rv^*/(k + rv^*)$, is continuous: it is 0 at λ^* and rises from there. (b) $\lambda^* \propto \kappa$; at α = 0 nuisance is free. (c) For $d \leq k$ there is no threshold. (d) *Linearity is needed.* If the encoder may also output $d - k$ unit-variance functions of s that are uncorrelated with s and with each other, then $J = 0$ is reached with no nuisance, which is below every linear minimum. In a nonlinear network, leakage above λ^* is therefore not forced by the objective: it reflects which zero-cost solution training finds. (3DShapes content takes 320 values, so such functions exist; numerically $J = 9{\times}10^{-7}$ against a linear minimum of 9.5 or more.)

Heuristic 1 (empty-direction traps). A direction u with $\Sigma u = 0$ gives the predictor no gradient along u. For cov, adding variance v there through a *fresh* signal (one uncorrelated with the current embedding) changes J by $v[c_p c(u) - 2\lambda] + \lambda v^2$, with $c(u) = \|Wu\|^2 - 2\rho\, u^\top Wu + 1$ ($\rho = 1$ for content, $1 - \alpha$ for nuisance). (Copying content that is already encoded does not help: its cross-covariance enters R, and the slope is 0.) If $Wu = 0$, the direction is filled only above $c_p/2$ for cov, *whatever* α is; the other arms have an analogous point set by their slope at a collapsed direction. Full local minimality is not proved. In gradient descent, 21/60 jointly trained and 6/135 fast-predictor runs ended in such states. So

the onset seen in training can lie above the global threshold $c_p\alpha(2-\alpha)/2$, up to about $c_p/2$ for cov (more if $c(u) > 1$); at $\alpha = 1$ the two coincide.

Theorem 2 (scale reduction; proved; constants numerical). Let R depend on z only through Σ. This holds in population for every Gaussian-target arm when z is Gaussian, and for the expected batch statistic at fixed batch size. In the linear model, z is Gaussian for every (A, B) only if s and n are, so for the kernel arms the theorem and the constants in Table 1 assume Gaussian s and n (cov needs no such assumption). 3DShapes content is discrete, so for the kernel arms these constants are approximations. Then the minimisers depend on (α, λ, c_p) only through $t = c_p\kappa/\lambda$, the nuisance variance of a minimiser is non-increasing in t, and the threshold is $\lambda^*_{\text{reg}} = c_p\kappa(\alpha)/t^*_{\text{reg}}$, where t^*_{reg} depends only on R, d, k, m. With a learned predictor, every such arm has $\lambda^* \propto \alpha(2-\alpha)$.

We computed population forms of each regulariser and validated them against the batch estimators by Monte Carlo. We then found t^* by bisection on the global eigenvalue minimum and confirmed it by gradient descent on (A, B) (Table 1). The expected cov penalty includes a finite-batch term, which gives $t^* = 1.952$ instead of 2. A second-order expansion of each regulariser at I ("curvature at I") predicts t^* badly: it is off by a factor of up to 5.3 at $d = 12$ and 12 for KSD at $d = 24$. **That proxy is refuted.** At $d = 12$, for every arm t^* agrees within 2% with the value set by the regulariser's slope at a collapsed direction. Above the threshold, only cov follows $v = 1 - \lambda^*/\lambda$. The kernel arms rise more slowly: at $\lambda = 2\lambda^*$, $v = 0.22$ (KSD), 0.40 (SIGReg) and 0.37 (MMD), against 0.5 (Figure 1 a).

VICReg (heuristic). Theorem 2′s reduction applies, but whether a threshold exists is not proved, and the variance term is a hinge. Content alone can satisfy it on all d coordinates by spreading its rank-k covariance as a tight frame, paying only the covariance term. In gradient descent no nuisance is bought up to $\lambda = 4.6$, and the nuisance share is 0.17 at $\lambda = 10$. We registered "≈ 7" as a low-confidence threshold.

Table 1: Linear-model thresholds λ^* (code units, $k = 6$, $m = 5$, $d = 12$ unless stated, learned linear predictor), from `theory/check_t2.py`. The last column is the ratio of the true t^* to its curvature-at-I approximation. At $d = 6$ no arm has a threshold.

arm	$\lambda^*, \alpha = 1$	$\alpha = 0.5$	$\alpha = 0.25$	$d = 24, \alpha = 1$	t^*/t_{quad}
cov	0.0427	0.0320	0.0187	0.0213	0.97
SIGReg	0.0389	0.0292	0.0170	0.0431	1.9
KSD	0.939	0.704	0.411	0.588	5.3
sliced KSD	7.33	5.50	3.21	8.04	2.3
MMD	17.1	12.8	7.48	18.0	2.3

Heuristic 2 (noise on the predictor input, T3). Suppose the predictor sees $z + \sigma\xi$, with ξ scaled by the batch standard deviation (study 1) or fixed. If content and nuisance occupy orthogonal principal directions (and, for batch scaling, coordinate-aligned ones), the per-unit-variance costs can be computed in closed form. Minimising over eigenvalues then gives these results. For cov, σ leaves λ^* unchanged ($0.0427 \longrightarrow 0.0424$ at $\sigma = 1$). For coupled regularisers, taxing content shrinks it, which makes filling the spare directions more valuable, so batch-scaled σ *lowers* λ^*: for KSD, 0.94 / 0.78 / 0.25 at $\sigma = 0$ / 0.5 / 1. Fixed-scale $\sigma = 0.5$ moves it less (0.88), and $\sigma = 1$ lowers it to 0.002. This predicts the sign of finding 2 but not its size: at study 1′s $\lambda = 0.2$ it predicts no leakage even at $\sigma = 1$. Minibatch KSD training does not satisfy the axis-alignment assumption, hence the label.

Proposition 3 (what a Gaussian target can identify; T4(i), proved). For KSD with an IMQ kernel, MMD with a Gaussian kernel, sliced KSD and SIGReg with random directions, all targeting $\mathcal{N}(0, I)$, the regulariser is unchanged when z is replaced by Oz for any orthogonal O (for the sliced arms, in expectation over the random directions). With a final linear encoder layer, a predictor $W \mapsto OWO^\top$, and predictor-input noise that is absent or isotropic with fixed scale (batch-scaled noise, $\sigma^2 \operatorname{diag}\Sigma$, is not isotropic; E5 uses $\sigma = 0$), the objective is $O(d)$-invariant, so its minimisers form unions of orbits. Orthogonally invariant metrics (probe R^2, R^2 after an orthogonal fit) are constant on an orbit. A permutation-level metric such as the mean correlation coefficient (MCC) is not fixed by the objective. A matched-prior arm whose minimiser lies in such an orbit cannot be told apart *on orthogonal-fit* R^2. **T4(ii)** is Comon's theorem [9, Thm 11]: if s has independent non-Gaussian components (at most one Gaussian) and $z = Ms$ with orthogonal M has independent components, then M is a signed permutation. So if a matched-prior arm's minimiser lies in the same orbit as a Gaussian arm's, its effect can show only on a permutation-level metric such as MCC. The proposition does not exclude that a matched prior moves training to a different orbit, which can change orthogonal-fit R^2 as well. This needs independent scalar factors. Study 1′s hue is coded as (cos, sin), a dependent pair whose law is nearly rotation-invariant; scale is the one factor that fits.

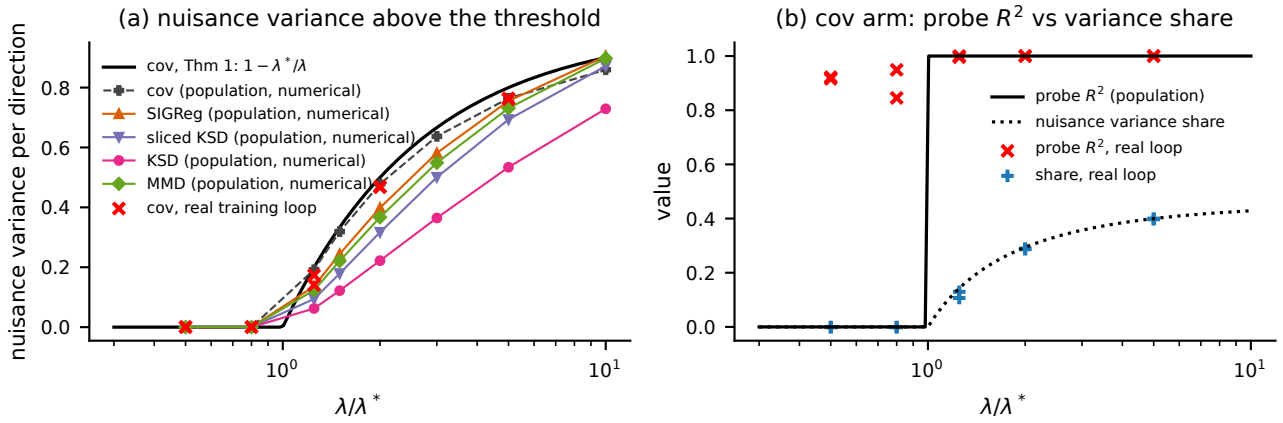


Figure 1: Linear model (cov arm unless stated; $d = 12$, $\alpha = 1$). (a) Nuisance variance per nuisance direction against λ/λ^*: Theorem 1′s curve, the population optimum of each regulariser (Theorem 2, numerical), and the real training loop (`linear` encoder, cov arm, synthetic linear-Gaussian views, 2 seeds). (b) Population probe R^2 steps; the variance share is continuous. In the real loop the probe already reads 0.85–0.95 below the threshold from residual nuisance variance $\leq 4.2\times10^{-4}$: a scale-invariant probe detects any nonzero variance.

Theory meets code. The real training loop, with the `linear` encoder, linear predictor and cov arm on synthetic linear-Gaussian views, reproduces Theorem 1. The onset lies between $0.8\lambda^*$ and $1.25\lambda^*$ in both seeds, and per-direction nuisance variance is 0.467/0.469 at $2\lambda^*$ and 0.761/0.764 at $5\lambda^*$ against the finite-batch optimum 0.466 and 0.760 (9/10 runs within 0.03; the other run has one trapped empty direction, as in Heuristic 1). At $d = k = 6$ the same loop does *not* reach the global minimiser of Theorem 1(a): at $5\lambda^*$ and $20\lambda^*$ (λ^* of $d = 12$) it ends in 5/6 and 6/6 seeds with nuisance in place of one content direction (leakage 0.19–0.36, content R^2 0.70–0.84). We had written this check as a gate; it failed, and we report it as a non-global state that training reaches.

3 Predictions and experimental design

The base setting is study 1′s: 3DShapes, a convolutional encoder with an MLP predictor, $d = 12$, $\alpha = 1$, $\sigma = 0$, uniform content law, batch 256, 2000 steps, held-out seeds 1–5. Cached study-1 runs were reused where the configuration matched. There is no λ tuning: each arm is run on about seven log-spaced λ levels bracketing its linear-model threshold. New metrics are the content and nuisance *variance shares* (the fraction of embedding variance explained by OLS on each code), per-principal-direction nuisance variance, and MCC (Hungarian matching on absolute correlations). The new arm `cov` is $\left\|\hat{\Sigma} - I\right\|_F^2$, and the new encoder `linear` is one linear map from pixels. All intervals are 95% seed-bootstrap intervals (2000 draws).

Before any study-2 network was trained, we froze `theory/predictions.md` (P1–P18) and the threshold estimator. $\hat{\lambda}^*$ is the smallest λ at which the 5-seed mean linear leakage exceeds $\tau = 0.10$, interpolated in log λ, with a seed bootstrap, and reported as censored ("<") when the crossing lies outside the grid. The experiments are E1 (λ paths for six arms × $\alpha \in \{0.25, 0.5, 1\}$ at $d = 12$, and $\alpha = 1$ at $d \in \{6, 24\}$), E2 (the linear-encoder rung), E3 (2000 vs 8000 steps), E4 (batch- vs fixed-scale σ) and E5 (MCC for study 1′s identifiability cells). E6, an optional theory-guided arm, was cut for compute, following the registered cut order. Table 2 lists every prediction and its outcome.

Table 2: Registered predictions (`theory/predictions.md`, frozen in P0) and their outcomes. The verdict follows the registered criterion; the statistic is from `results2/verdicts.json`.

#	prediction	verdict	key statistic
P1	step-like onset	refuted (cov, VICReg); step holds for kernel arms	kernel arms rise within 0.30–0.60 decades; cov, VICReg ≥ 0.10 at lowest λ
P2	location within ×3	refuted	KSD 0.32× [0.29, 0.37], sliced KSD 0.29× [0.22, 0.38], SIGReg 0.72×, MMD 0.79×
P3	ordering SIGReg ≈ cov < KSD < sKSD < MMD	confirmed (cov censored)	successive ratios 10.5, 7.3, 6.3; CIs exclude 1
P4	α ratio in $[\alpha(2-\alpha), 1]$	confirmed at $\alpha = 0.25$ (4 kernel arms); 2 above 1 at $\alpha = 0.5$	ratio at $\alpha = 0.25$: 0.50–0.65 ($\propto \alpha$: 0.25)
P5	no threshold at d = 6	refuted	all 6 arms reach ≥ 0.10 at d = 6 (max 0.33–0.67)
P6	d = 24 ratios	not refuted; point values miss	KSD 0.36, SIGReg 0.76, MMD 0.81
P7	probe steeper than share	inconclusive	probe narrower in 3/6 arms
P8	linear rung within ×3	refuted (cov location and α ratio)	cov 9.4×, KSD 1.33× theory; α ratio cov 0.69, KSD 1.06
P9	cov form $1 - \lambda^*/\lambda$	cov not refuted (one range met per rung); kernel part confirmed	linear-rung cov v = 0.30 / 0.53 at 2/5 λ^*
P10	KSD/cov > 5	refuted (linear rung); consistent (conv)	linear rung 3.1; conv > 14.8 (cov censored)
P11	leakage persists at 8000 steps	refuted	8000 − 2000 steps: −0.36, −0.41, −0.33, −0.18
P12	spare directions (descr.)	–	unexplained share rises; kNN content $R^2 \longrightarrow 1.00$
P13	sign of batch σ (KSD)	confirmed (leakage)	KSD batch $\sigma = 1$: +0.66, +0.65
P14	size of σ effect (KSD)	refuted in size	$\lambda = 0.2$, $\sigma = 1$: +0.65 (predicted 0)
P15	fixed vs batch σ	refuted (KSD)	fixed − batch at σ 0.5: +0.13, +0.15
P16	no orth-fit R^2 gain from matched prior	not refuted	uniform: −0.02, +0.03 (CIs include 0)
P17	no hue-pair MCC gain	refuted	hue-pair MCC +0.25 at uniform λ 0.2
P18	top-k cov arm	not tested	E6 cut

4 Results

4.1 Thresholds: order and α-scaling hold, locations and $d = k$ fail (E1, E2)

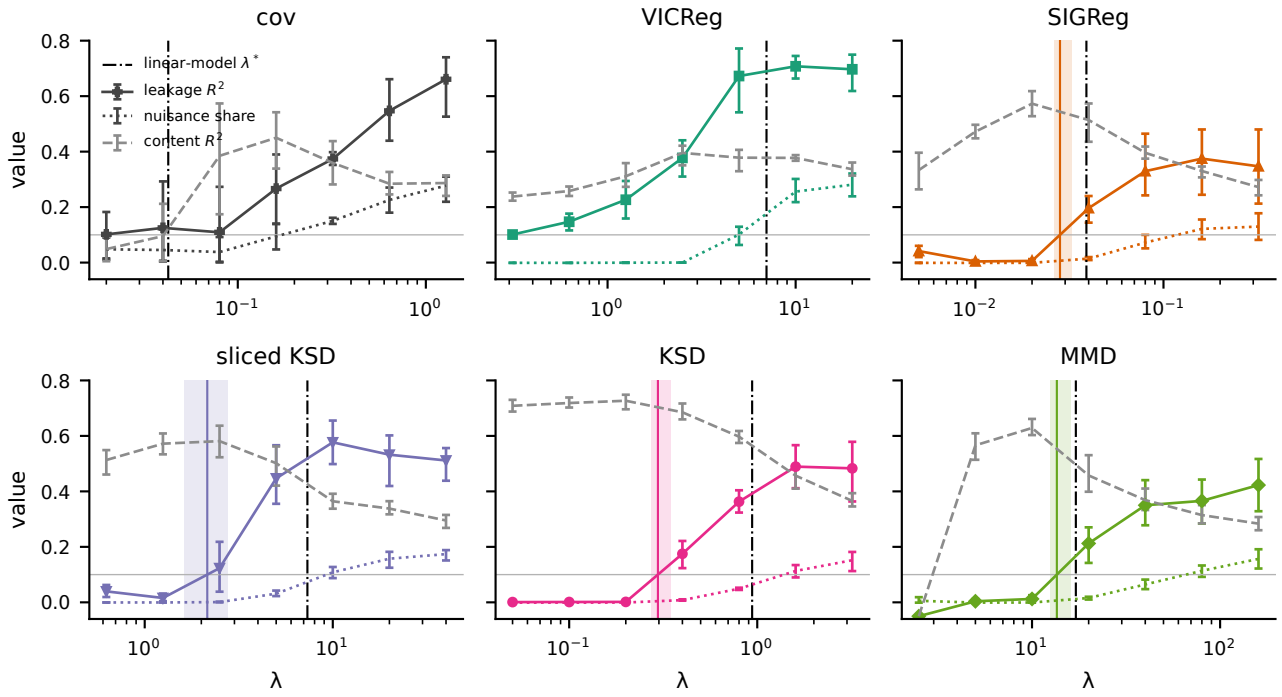


Figure 2: λ paths at the base setting (conv encoder, $d = 12$, α = 1). Lines show linear leakage R^2 (markers), nuisance variance share (dotted) and content R^2 (grey dashed); bars are 95% seed CIs. The dash-dotted line is the linear-model λ*, and the coloured band is $\hat{\lambda}^*$ with its CI. MMD's lowest λ (mean leakage −7.74, a probe failure on collapsed seeds) is clipped at −0.05.

Table 3: Measured thresholds (linear leakage, τ = 0.10). "Ratio" is $\hat{\lambda}^*$ over the linear model; the α and $d = 24$ columns are $\hat{\lambda}^*$ relative to α = 1, $d = 12$. "–" means censored. CIs for the ratios are in the text.

arm	linear model λ^*	$\hat{\lambda}^*$ [95% CI]	ratio	$\alpha = 0.5$	$\alpha = 0.25$	$d = 24$	$\hat{\lambda}^*$ at $d = 6$
cov	0.043	< 0.020	–	–	–	–	0.362
VICReg	≈ 7.0	< 0.31	–	–	–	–	< 2.5
SIGReg	0.039	0.0282 [0.0262, 0.0327]	0.72	0.92	0.52	0.76	0.0412
sliced KSD	7.3	2.15 [1.62, 2.78]	0.29	1.20	0.65	–	5.72
KSD	0.94	0.296 [0.272, 0.345]	0.32	1.27	0.55	0.36	1.24
MMD	17	13.6 [12.5, 16.0]	0.79	0.94	0.50	0.81	< 20

The kernel arms step; cov and VICReg start in a collapse regime (P1). SIGReg, sliced KSD, KSD and MMD rise from below 0.10 to half their maximum within 0.30–0.60 decades of λ (Figure 2). Cov and VICReg do not: at their lowest λ, cov has embedding standard deviation 0.000 and content R^2 0.05, and VICReg has effective rank 2.4. Yet the probe still reads leakage of 0.102 and 0.101 from these near-collapsed embeddings, as the scale invariance of Corollary 1(a) allows (Figure 1 b). Their thresholds are censored, and P1 fails by its registered criterion, by a margin of at most 0.002, on near-collapsed embeddings.

Order confirmed; locations early (P2, P3). The measured thresholds order as Theorem 2′s constants: SIGReg < KSD < sliced KSD < MMD. The successive ratios are 10.5 [8.9, 12.5], 7.3 [5.3, 9.6] and 6.3 [4.8, 8.8], and all CIs exclude 1. Locations are 0.29–0.79 times the linear model. SIGReg and MMD fall within the registered factor 3, while KSD (0.32) and sliced KSD (0.29) fall just outside it (both CIs, [0.29, 0.37] and [0.22, 0.38], include 1/3), so P2 is refuted on the point estimates. Every uncensored conv-encoder threshold is *earlier* than the linear model.

α scaling: not proportional to α (P4). At α = 0.25 the ratio $\hat{\lambda}^*(\alpha)/\hat{\lambda}^*(1)$ is 0.50–0.65 across the four kernel arms, with every CI above 0.41. That is far from the 0.25 that $\lambda^* \propto \alpha$ requires, and close to the learned-predictor value 0.44, though not all CIs include it (Figure 3 a; SIGReg 0.52 [0.43, 0.70], KSD 0.55 [0.44, 0.75], sliced KSD 0.65 [0.50, 0.86], MMD 0.50

[0.41, 0.63]). At α = 0.5, KSD (1.27 [0.94, 1.57]) and sliced KSD (1.20 [0.77, 1.65]) have point estimates above 1, the bound set by the trap mechanism (Heuristic 1′s escape point does not depend on α and equals the threshold at α = 1), though their CIs include 1; for these two arms the point estimates are not monotone in α.

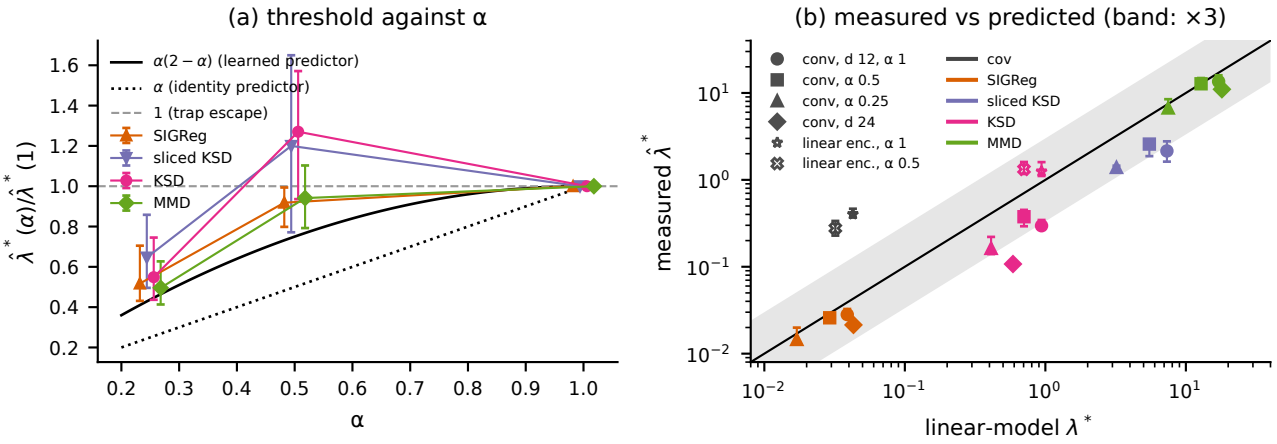


Figure 3: (a) Threshold ratio against α for the four kernel arms, with α(2−α) (Theorem 1, learned predictor), α (identity predictor) and 1 (trap escape, Heuristic 1). (b) Measured against predicted thresholds for every uncensored setting; the grey band is the registered factor 3. Open markers: linear pixel encoder.

$d = k$ is not threshold-free (P5, refuted). At $d = 6$ all six arms reach leakage ≥ 0.10 (maxima 0.33–0.67). Leakage is lower than at $d = 12$ at every shared λ, and onsets move up (KSD 1.24 vs 0.296; cov 0.362). Content R^2 falls as leakage rises: the network trades content for nuisance. The global minimiser of the linear model never does this, but the same training loop on the linear model does (Theory meets code), so the $d = k$ leakage need not come from nonlinearity: the same training dynamics produce it in the linear model.

$d = 24$ (P6). Wider embeddings lower the thresholds by more than predicted: the ratio is 0.36 [0.31, 0.40] for KSD (predicted ≈ 0.63) and 0.76 [censored, 0.83] for SIGReg and 0.81 [0.64, 0.90] for MMD (predicted 0.9–1.2). The registered "against" criterion (< 0.6) is not met, but the point predictions miss.

Probe vs share (P7, inconclusive). The 10–90% rise is narrower for the probe than for the share in 3 of 6 arms, by small margins (probe-failure levels excluded). The share crosses 0.05 two to three times later than the probe crosses 0.10 (KSD 0.808 vs 0.296), so even a small nuisance variance is decodable.

Linear encoder and functional form (P8–P10). With one linear map from pixels, KSD's threshold (1.25 [1.11, 1.60]) is 1.33× the linear model, but cov's (0.403 [0.368, 0.465]) is 9.44×, and its α ratio is 0.69 [0.55, 0.87]. Content is not a linear function of pixels, so content is not free for a pixel-linear map, and Theorem 1′s premise fails at this rung. The KSD/cov ratio is 3.1 [2.6, 4.0] instead of > 5 (P10 refuted at this rung; at the conv encoder it is > 14.8, cov censored, consistent with P10). Above the threshold, the linear-rung cov's per-direction ratio is 0.30 at $2\hat{\lambda}^*$ and 0.53 at $5\hat{\lambda}^*$ (theory 0.5 and 0.8). The conv-encoder cov reaches 2.23 at λ = 1.28, i.e. nuisance directions carry *more* variance than content directions, which is impossible in the linear model. The kernel arms rise more slowly than $1 - \lambda^*/\lambda$, as predicted (KSD 0.11, SIGReg 0.21, MMD 0.18 at $2\hat{\lambda}^*$).

4.2 Frontiers: the ranking changes (not predicted)

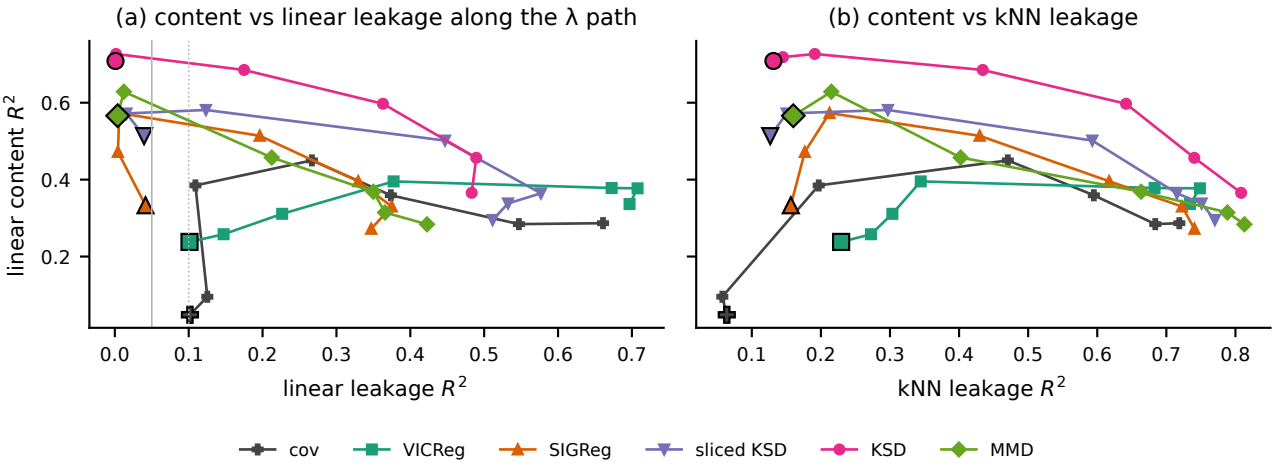


Figure 4: Content–leakage paths along λ at the base setting (black-edged marker: lowest λ shown; MMD's λ = 2.5 is omitted, see Figure 2). Up and to the left is better.

The theory places thresholds, not frontiers, so this comparison was not predicted. Figure 4 compares arms on the frontier instead of at one tuned λ. The best content R^2 reachable with leakage ≤ 0.05 is 0.73 [0.72, 0.75] for KSD, 0.63 [0.60, 0.66] for MMD, 0.57 [0.53, 0.62] for SIGReg and 0.57 [0.53, 0.64] for sliced KSD. Cov has no level with leakage ≤ 0.10 (best 0.38 [0.04, 0.57] at ≤ 0.20). VICReg has none ≤ 0.05, and its whole path stays at content R^2 ≤ 0.40. Study 1 had found SIGReg leaking more than VICReg at VICReg's least-leaky level. On the frontier, VICReg comes last. Above the thresholds the order changes again. At matched content R^2 = 0.5, leakage is 0.16 [0.14, 0.19] for MMD and 0.21 [0.10, 0.27] for SIGReg, but 0.45 [0.40, 0.49] for KSD. Which arm "leaks least" depends on the operating point, not on the target law.

4.3 Linear leakage is a transient of 2000-step training (E3)

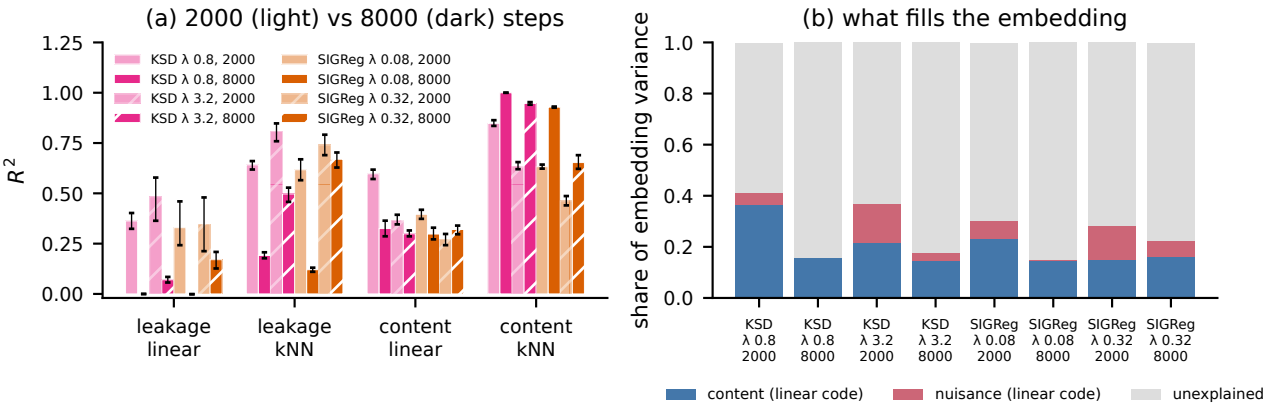


Figure 5: E3: KSD (λ 0.8, 3.2) and SIGReg (λ 0.08, 0.32), above their thresholds, at 2000 and 8000 steps. (a) Linear and kNN probes for leakage and content; (b) shares of embedding variance explained by the linear content and nuisance codes, and the unexplained rest.

Prediction P11, a weak one derived from the trap mechanism, was that leakage would persist. It is refuted in all four cells. From 2000 to 8000 steps, leakage changes by −0.36 [−0.40, −0.32] (KSD 0.8), −0.41 [−0.50, −0.30] (KSD 3.2), −0.33 [−0.47, −0.24] (SIGReg 0.08) and −0.18 [−0.32, −0.03] (SIGReg 0.32). At KSD 0.8 it falls to 0.00. Figure 5 b shows what replaces it. The linearly unexplained share rises from 0.59 to 0.84, and linear content R^2 falls (0.60 ⟶ 0.32), while kNN content R^2 rises to 1.00 and kNN leakage falls (0.64 ⟶ 0.19). At the higher λ levels kNN leakage stays high (KSD 3.2: 0.50; SIGReg 0.32: 0.67 at 8000 steps): there nuisance becomes less linearly accessible rather than absent. In the lower-λ cells the network fills the spare directions with *nonlinear functions of content*. This is the mechanism of Corollary 1(d), which lies outside the linear model; the corollary proves it zero-cost for cov only, and we have not shown that it is cheaper for KSD or SIGReg, whose targets a function of discrete content does not match exactly. The threshold therefore describes the regime that 2000-step training reaches, not an equilibrium of the network.

4.4 Noise on the predictor input (E4)

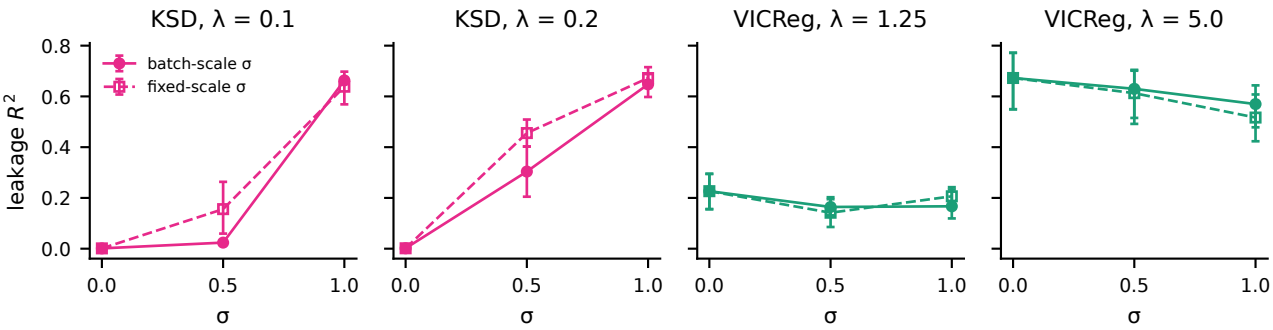


Figure 6: E4: linear leakage against σ for batch-scaled (filled) and fixed-scale (open) noise on the predictor input.

For KSD, batch-scaled σ raises leakage at both levels (P13): at λ = 0.1, σ = 1 adds +0.662 [0.650, 0.672] and σ = 0.5 adds +0.023 [0.016, 0.031]. That matches the registered pattern of an onset at σ = 1 but not at σ = 0.5 (P14, registered at $\hat{\lambda}^*(0)/2$; λ = 0.1 is about a third of $\hat{\lambda}^*(0)$). P14′s threshold ratios $\lambda^*(\sigma)/\lambda^*(0)$ were not measured: E4 has no λ path under σ. At λ = 0.2 the linear model predicts no leakage, yet σ = 1 adds +0.646 [0.596, 0.687], so the size is wrong (P14). The content share does not fall as predicted (0.44 ⟶ 0.52 ⟶ 0.51). Fixed-scale σ = 0.5 raises leakage *more* than batch-scaled σ = 0.5 (fixed − batch +0.13 [0.04, 0.24] at λ 0.1 and +0.15 [0.04, 0.26] at 0.2), the reverse of P15. Under fixed σ = 1, KSD shrinks the embedding (standard deviation 0.80 ⟶ 0.41). σ does not raise VICReg's leakage: all 8 of 8 contrasts with σ = 0 are negative (−0.16 to −0.02), with CIs that include 0, and there is no inflation under fixed noise (fixed − batch standard deviation +0.03 [0.00, 0.06]). The theory gets the sign right for KSD and nothing about VICReg.

4.5 Identifiability with the right metric (E5)

Table 4: E5: matched-prior KSD minus Gaussian-target KSD, mean [95% CI] over 5 seeds. Laplace pairs use study 1′s selected levels (Gaussian 0.2 vs matched 1.0) and both divided by 5.

law, λ (Gaussian vs matched)	orth-fit R^2	MCC	MCC hue pair	MCC scale	leakage	content R^2
uniform, λ 0.2 vs 0.2	−0.02 [−0.08, 0.04]	+0.09 [0.02, 0.14]	+0.25 [0.18, 0.32]	+0.22 [0.11, 0.33]	+0.12 [0.04, 0.25]	−0.06 [−0.11, −0.02]
uniform, λ 0.04 vs 0.04	+0.03 [−0.05, 0.11]	−0.01 [−0.05, 0.03]	+0.08 [−0.01, 0.18]	+0.06 [−0.01, 0.13]	0.00 [0.00, 0.00]	−0.02 [−0.07, 0.03]
Laplace, λ 0.2 vs 1.0	−0.50 [−0.56, −0.43]	−0.21 [−0.26, −0.17]	+0.12 [0.04, 0.22]	−0.61 [−0.69, −0.51]	+0.33 [0.26, 0.40]	−0.49 [−0.54, −0.45]
Laplace, λ 0.04 vs 0.2	−0.44 [−0.51, −0.38]	−0.24 [−0.30, −0.17]	−0.07 [−0.16, 0.02]	−0.63 [−0.76, −0.49]	+0.47 [0.41, 0.53]	−0.48 [−0.54, −0.40]

Orthogonal-fit R^2 shows no gain from the matched prior (P16 not refuted; Table 4): the uniform-law differences have CIs that include 0. At the Laplace law the matched arm loses content and leaks, and its orthogonal-fit R^2 is lower. MCC tells a different story. At the uniform law and λ = 0.2, the matched prior raises MCC by +0.09 [0.02, 0.14] overall and by +0.22 [0.11, 0.33] on scale, the factor where T4 fits. It also raises MCC by +0.25 [0.18, 0.32] on the hue pair, which P17 said it would not. This is consistent with the metric argument of Proposition 3: MCC sees what orthogonal-fit R^2 does not. But the gain is not confined to the independent scalar factor, it vanishes at λ = 0.04, and it comes with more leakage (+0.12 [0.04, 0.25]) and less content (−0.06 [−0.11, −0.02]).

5 What the theory explains, and what it does not

Finding 1 is explained in form. Kernel arms have threshold-like onsets, their order follows the regulariser's slope at a collapsed direction, their α ratios lie above α(2−α), near 1 or (two arms at α = 0.5) above it, and not ∝ α, and they fall with d. The details are not explained: conv-encoder thresholds come at 0.29–0.79 times the prediction (always early), $d = k$ still leaks, and a linear pixel encoder moves cov's threshold up by 9.44×. Above all, the threshold is not an equilibrium. E3 shows the network leaving the nuisance solution for nonlinear content features, the possibility Corollary 1(d) leaves open (proved free for cov, not for the kernel arms). The theory says what the nonlinear network can do at zero cost (for cov), but not why 2000-step training first takes the nuisance route; simplicity bias [10] is a candidate we did not test. *Finding 2* is explained in sign for KSD only. The size is wrong, the fixed/batch ordering is wrong, and σ does not raise VICReg's leakage in this grid. *Finding 3* is explained as a metric artefact, in part. Within one orbit, orthogonal-fit R^2 cannot show a matched-prior gain, and in the data it shows none. MCC does show one at the uniform law, though not only in the cell

the theory singled out; at the Laplace law the matched prior lowers both metrics. The theory is silent on kNN leakage, which is already 0.13–0.22 for the kernel arms at levels where linear leakage is below 0.05, on content levels, and on the low-λ collapse that hides cov's and VICReg's thresholds.

6 Related theory

Linear models of self-supervised learning are well studied. Tian et al. analyse the dynamics of non-contrastive learning with a learned predictor, including eigenmode growth thresholds and the alignment of the predictor's eigenbasis [11]. Our Lemma 1(ii) and Heuristic 1 are close in spirit, but that work has no content/nuisance split, no α and no distributional regulariser. Jing et al. explain dimensional collapse in contrastive learning by augmentation strength [12]. Simon et al. derive stepwise eigenmode learning [13], and Balestriero & LeCun cast VICReg and related losses as spectral embeddings with closed-form optima [14]. Our eigenvalue reduction (Theorems 1–2) shares this spectral view. Shwartz-Ziv et al. give an information-theoretic reading of VICReg [15]. For three of these papers we read only the abstract and a keyword scan, so "not found" is limited to what we read. In the papers we examined, we did not find the λ threshold, its α(2–α) form, the collapse of unfillable directions, the scale reduction for Σ-dependent regularisers, or the per-regulariser constants. These are elementary consequences of standard matrix inequalities, and we claim them only as derivations and tests in this setting. T4(ii) is Comon's theorem [9]. Zimmermann et al. state the same orthogonal-versus-permutation dichotomy for contrastive learning, measured by R^2 and MCC [16]. T4 is an analogue of their result, with the asymmetry placed in the regulariser's target law. Von Kügelgen et al. isolate content when the encoder dimension equals the content dimension [17], the $d = k$ case of Theorem 1(a). Whether KSD with our surrogate scores vanishes only at the target [18] is assumed, not verified.

7 Limitations and reproducibility

The theory is a linear model of a nonlinear network, and E3 shows where that matters. All experiments use one rendered dataset with six factors, small encoders, 2000 steps (8000 in E3), and 5 seeds. Two arms' thresholds are censored by a low-λ collapse that the frozen estimator cannot see past. T2′s constants are numerical, and T3 and the VICReg analysis are heuristic. E6 was not run. The references follow the appendix. Every number in this paper is generated from `results2/` (and `results/` for study 1) by `stein/report2.py`, and the reproduction steps are in `REPRODUCE2.md`. An AI agent (Claude) ran the experiments and drafted the text and the proofs. The proofs were checked numerically (gradient descent against closed forms, `theory/check_t*.py`), not formally.

Appendix A. Proofs

Notation as in Section 2. $C := \mathrm{Cov}(z, z') = P + (1-\alpha)Q = \Sigma - \alpha Q$. Every pair of PSD matrices (P, Q) with rank $P \le k$ and rank $Q \le m$ is realisable by some (A, B).

Proof of Lemma 1. (i) $z - z' = B(n - n')$ and $\mathrm{Cov}(n - n') = 2\alpha I$. (ii) $\mathbb{E}\|Wz - z'\|^2 = \mathrm{tr}(W\Sigma W^\top) - 2\,\mathrm{tr}(WC) + \mathrm{tr}\,\Sigma$ is a convex quadratic in W, minimised by any W with $W\Sigma = C$, e.g. $W = C\Sigma^+$ (range $C \subseteq$ range Σ). The minimum is $\mathrm{tr}\,\Sigma - \mathrm{tr}(C\Sigma^+C)$. With $C = \Sigma - \alpha Q$, $\Sigma\Sigma^+\Sigma = \Sigma$ and range $Q \subseteq$ range Σ, we get $\mathrm{tr}(C\Sigma^+C) = \mathrm{tr}\,\Sigma - 2\alpha\,\mathrm{tr}\,Q + \alpha^2\,\mathrm{tr}(Q\Sigma^+Q)$. Let $\tilde{Q} = \Sigma^{+1/2}Q\Sigma^{+1/2}$ and let Π be the projector onto range Σ. From $0 \preccurlyeq Q \preccurlyeq \Sigma$ we get $0 \preccurlyeq \tilde{Q} \preccurlyeq \Pi$, so $\tilde{Q}^2 \preccurlyeq \tilde{Q}$. Since $Q = \Sigma^{1/2}\tilde{Q}\Sigma^{1/2}$, $\mathrm{tr}(Q\Sigma^+Q) = \mathrm{tr}(\Sigma^{1/2}\tilde{Q}^2\Sigma^{1/2}) \le \mathrm{tr}(\Sigma^{1/2}\tilde{Q}\Sigma^{1/2}) = \mathrm{tr}\,Q$, which gives the bound. Equality forces $\Sigma^{1/2}(\tilde{Q} - \tilde{Q}^2)\Sigma^{1/2} = 0$, i.e. $\tilde{Q} = \tilde{Q}^2$ on range Σ. Then $\tilde{P} := \Pi - \tilde{Q}$ is the complementary projector, and $\tilde{P}\tilde{Q} = 0 \Leftrightarrow P\Sigma^+Q = 0$. ∎

Proof of Theorem 1. By Lemma 1, $J \ge J_b := c_p\kappa\,\mathrm{tr}\,Q + \lambda R(\Sigma)$, with equality for the identity predictor. Fix Σ and let $\sigma_1 \ge \ldots \ge \sigma_d$ be its eigenvalues. Over decompositions $\Sigma = P + Q$ (PSD, rank $P \le k$, rank $Q \le m$) we have $\mathrm{tr}\,Q = \mathrm{tr}\,\Sigma - \mathrm{tr}\,P$. Since $P \preccurlyeq \Sigma$, Weyl's inequality gives $\lambda_i(P) \le \sigma_i$, so $\mathrm{tr}\,P \le \sum_{i\le k}\sigma_i$. This bound is attained by the spectral split (P = top-k eigen-part, Q = the rest), for which $\tilde{Q}$ is a projector, so $J = J_b$ there for both predictors. A decomposition exists only if rank $\Sigma \le k + m$, i.e. at most m of the σ_i with $i > k$ are nonzero. Hence $\min J = \min_\sigma f(\sigma)$ with $f(\sigma) = \sum_{i\le k}\lambda(\sigma_i - 1)^2 + \sum_{i>k}[c_p\kappa\sigma_i + \lambda(\sigma_i - 1)^2]$, subject to $\#\{i > k : \sigma_i > 0\} \le m$, $\sigma_i \ge 0$ and σ sorted. Each summand is strictly convex. The unconstrained minimiser ($\sigma_i = 1$ for $i \le k$, $\sigma_i = v^*$ for $i > k$) is sorted because $v^* < 1$. Under the cardinality constraint, a bottom slot set to v^* instead of 0 lowers f by $(\lambda - c_p\kappa/2)^2/\lambda > 0$ when $\lambda > \lambda^*$. So exactly r slots take v^* when $\lambda > \lambda^*$ and none otherwise, which fixes the spectrum of every minimiser. (If $d \le k$ there are no slots $i > k$: the spectrum is all ones, $\Sigma = I_d$, and any optimal decomposition minimises $\mathrm{tr}\,Q$; $\mathrm{tr}\,P \le d = \mathrm{tr}\,\Sigma$ is attained only

with $Q = 0$, so $B = 0$ and $P = I_d$, which is part (a).) For $d > k$ it remains to identify P. Since $\kappa > 0$, $v^* < 1$, so the top eigenvalue 1 has a k-dimensional eigenspace. Any optimal decomposition minimises $\operatorname{tr} Q$, so $\operatorname{tr} P = k$ with $\lambda_i(P) \le 1$ and rank $P \le k$, which forces k unit eigenvalues. For a unit eigenvector u of P, $u^\top \Sigma u \ge u^\top P u = 1 = \sigma_1$, so u lies in the eigenvalue-1 eigenspace. Hence $P = UU^\top$, and $Q = \Sigma - P = v^* V V^\top$ with $V \perp U$. For the learned predictor, any minimiser of J attains J_b (otherwise $J > \min J_b = \min J$), so the same structure holds. ∎

Proof of Corollary 1. (a) The probe R^2 for n_j is $(B^\top \Sigma^+ B)_{jj}$, so the mean over j is $\operatorname{tr}(\Sigma^+ B B^\top)/m = \operatorname{tr}((UU^\top + VV^\top/v^*)\, v^* V V^\top)/m = r/m$ when $v^* > 0$, and 0 when $B = 0$. The share is $\operatorname{tr} Q / \operatorname{tr} \Sigma = r v^*/(k + r v^*)$. (b), (c) follow from $\lambda^* = c_p \kappa/2$ and from $r = 0$ when $d \le k$. At $\alpha = 0$ (outside the model's range, taken as a limit), $\kappa = 0$, so $J = \lambda R$: the minimisers have $\min(d, k+m)$ unit eigenvalues and the rest 0, with any feasible split of Σ into P and Q; nuisance costs nothing. (When $d > k + m$, $P + Q = I$ is not reachable and $\min J = \lambda(d - k - m) > 0$.) (d) Take $z = (s, g(s))$ with g as stated. Then $\Sigma = I$, so $R = 0$, and $z = z'$, so the prediction loss is 0 and $J = 0$. For $d > k$ the linear minimum is $r[c_p \kappa v^* + \lambda(v^* - 1)^2] + \lambda(d - k - r) > 0$. With the identity predictor, $J = 0$ forces $\mathbb{E}\|z - z'\|^2 = 0$, so z does not depend on the resampled nuisance. ∎

Proof of Theorem 2. The proof of Theorem 1 up to "$\min J = \min_\sigma$" used only that R depends on Σ. Hence $\min J = \min_\Sigma [c_p \kappa N(\Sigma) + \lambda R(\Sigma)] = \lambda \min_\Sigma [t N(\Sigma) + R(\Sigma)]$, where $N(\Sigma) = \sum_{i>k} \sigma_i$ is the nuisance variance of the spectral split and Σ ranges over PSD matrices of rank $\le k + m$. The minimiser set depends on t only. If Σ_1 is optimal at t_1 and Σ_2 at $t_2 > t_1$, then adding $t_1 N_1 + R_1 \le t_1 N_2 + R_2$ and $t_2 N_2 + R_2 \le t_2 N_1 + R_1$ gives $(t_2 - t_1)(N_2 - N_1) \le 0$, so N is non-increasing in t. Let t^* be the boundary between minimisers with $N > 0$ and with $N = 0$. Then $\lambda^* = c_p \kappa / t^*$. ∎ That t^* exists, is positive and finite, and its value are numerical (Table 1).

Proof of Proposition 3. The IMQ and Gaussian kernels are functions of $\|x - y\|$. The score of $\mathcal{N}(0, I)$, $-x$, is equivariant, so every term of the Stein kernel is built from inner products of x, y and $x - y$, and $u(Ox, Oy) = u(x, y)$. MMD's target is invariant. The sliced statistics average over Haar-distributed directions, whose law is preserved by $g \mapsto O^\top g$, so they are invariant in expectation over the directions. Without noise, or with isotropic noise $\sigma\xi$ of fixed scale (whose law is rotation-invariant), the prediction loss is invariant under $z \mapsto Oz$, $W \mapsto OWO^\top$; an MLP predictor absorbs $O^\top$ and O in its first and last linear layers, and the encoder's last linear layer absorbs O. ∎

Numerical checks. `theory/check_t1.py` compares gradient descent with Theorem 1 for five shapes (d, k, m) (including $d \le k$ and $m < d - k$), $\alpha \in \{0.25, 0.5, 1\}$ and $\lambda/\lambda^* \in \{0.5, 0.8, 1.25, 2, 5\}$, with both predictors and in minibatch training. `check_t2.py` validates the population regularisers by Monte Carlo and checks t^* by gradient descent. `check_t4.py` finds rotation changes of the Gaussian-target estimators of $|z| \le 0.7$ standard errors for sliced KSD, SIGReg and MMD; KSD is invariant sample by sample (relative change 1×10^{-7}, floating-point error), against $z = 15.8$ for the uniform-prior KSD. For three independent sources, the matched Laplace prior reaches MCC 0.88/1.00/1.00 over three seeds, while the uniform surrogate stalls at a rotated solution (MCC 0.81–0.86) whose KSD (0.019) exceeds that of the signed permutation (0.0045): the population claim holds, but training misses it. A Gaussian target gives MCC 0.76–0.91, i.e. an arbitrary rotation. All results are in `results2/theory_t*.json`.

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